Practical Assignment (problem solving questions)

Level: Year 2 – Medical Students

1- **100 people are tested for the disease. 15 people have the disease; 85 people are not diseased. 50 patients were positive for the test. Among 85 individual found not to be diseased according to the gold standard, 40 individual were found to be positive for the test, calculate Prevalence of Disease, Sensitivity of the test & Specificity of the test.**

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2- **Many of the students at the boarding school, including 6 just coming down with varicella, went home during the Thanksgiving break. About 2 weeks later, 4 siblings of these 6 students (out of a total of 10 siblings) developed varicella. Calculate the secondary attack rate among siblings .**

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3- **A new screening test was evaluated for Disease A among 500 individuals using a gold standard.**

- **According to the gold standard, 80 individuals had Disease A.**
- **The screening test was positive in 60 diseased individuals.**
- **Among the 420 non-diseased individuals, 90 tested positive.**

1. **Calculate the prevalence of Disease A.**
2. **Construct a 2 × 2 table.**
3. **Calculate:**

- **Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV)**

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4- In a community screening program:

- **Disease prevalence = 8%**
- **Total population screened = 2,500**
- **The screening test correctly identified 160 true cases.**
- **2,070 individuals were correctly identified as disease-free.**

Questions:

1. **Calculate the number of diseased and non-diseased individuals.**
2. **Complete the 2 × 2 table.**
3. **Calculate sensitivity, specificity, PPV, and NPV.**
4. **Based on PPV and NPV, state whether the test is more suitable for screening or diagnosis.**

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5- **In City B, the following data were recorded in 2023:**

- **Total population: 90,000**
- **Women aged 15–49 years: 24,000**
- **Married women aged 15–49 years: 18,000**
- **Total live births: 2,700**

Age group (years)	Live births	Women
15–19	135	5,000
20–24	720	6,000
25–29	810	5,000
30–34	540	4,000
35–39	315	2,500
40–44	135	1,200
45–49	45	300

1. **Calculate the Crude Birth Rate (CBR).**
2. **Calculate the General Fertility Rate (GFR).**
3. **Calculate ASFRs.**
4. **Calculate the Total Fertility Rate (TFR).**

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6- **A health survey in a town with a population of 40,000 revealed:**

- **Existing cases of hypertension at start of year: 3,200**
- **New hypertension cases during the year: 400**
- **Population at risk: 36,800**

- **Total episodes of acute respiratory infection (ARI): 2,400**
- **Number of persons affected by ARI: 2,000**
- **Total days lost due to ARI: 4,800**
- **Population screened for anemia: 3,000**
- **Anemia cases detected: 450**

1. **Calculate point prevalence of hypertension.**
2. **Calculate period prevalence of hypertension.**
3. **Calculate incidence rate of hypertension.**
4. **Calculate attack rate of ARI.**
5. **Calculate prevalence of anemia.**

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A depression screening test is compared blindly with a gold standard in 200 psychiatric outpatients. Of 50 true depressed patients, 35 tested positive; of 150 non-depressed individuals, 30 tested positive. calculate the following:

1. Sensitivity
2. Specificity
3. Positive Predictive Value (PPV)
4. Negative Predictive Value (NPV)
5. Prevalence rate of depression

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A sudden increase in cases of acute gastroenteritis is reported among students in a boarding school over 3 days. Many students present with vomiting, diarrhea, and abdominal cramps, describe the **steps you would follow to investigate this epidemic**.

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In a table form **Compare between a common source outbreak and a propagated epidemic** in terms

1. Source of infection
2. Mode of transmission
3. Epidemic curve pattern
4. Incubation period distribution
5. Duration of outbreak
6. Main control measures

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During a wedding party, several attendees developed acute gastrointestinal illness characterized by vomiting and diarrhea within 12 hours of the event. A cohort study was conducted to identify the food item responsible for the outbreak.

The following data were collected:

Food Item	Ate Food & Became Ill	Ate Food & Not Ill	Did Not Eat & Became Ill	Did Not Eat & Not Ill
Chicken	40	20	10	30
Salad	35	25	15	25
Cake	50	10	5	40

Questions

1. Calculate the **Relative Risk (RR)** for each food item.
2. Interpret the RR values.
3. Identify the **most likely food item responsible** for the outbreak.
4. Justify your answer based on epidemiological principles.